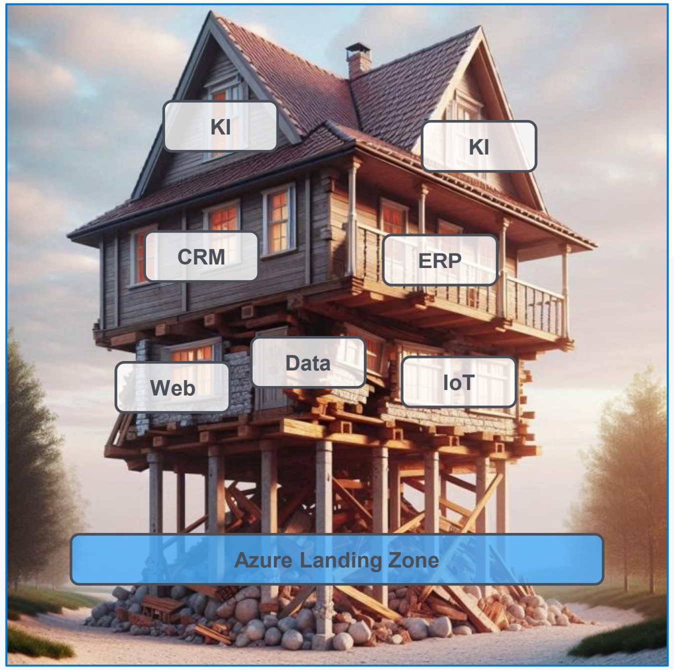
Success Stories 🏆

Building scalable and flexible **Azure Landing Zones**.

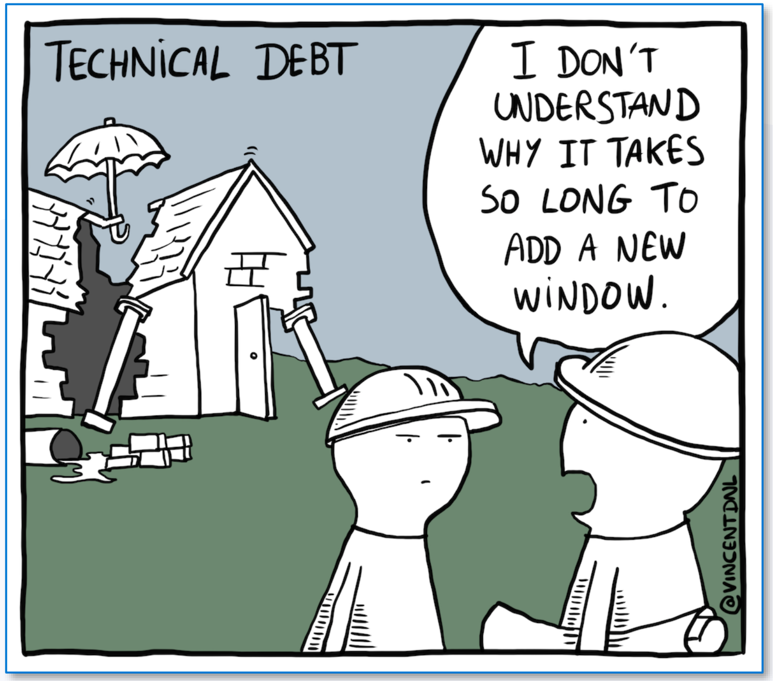
(by Stefan Rapp, 9th May 2025, 14:00 – 15:00)

Status quo – The “*usual*” way to the cloud

Avoid "*technical debt*" in the cloud



"*Landing Zone*" → **Groundwork**

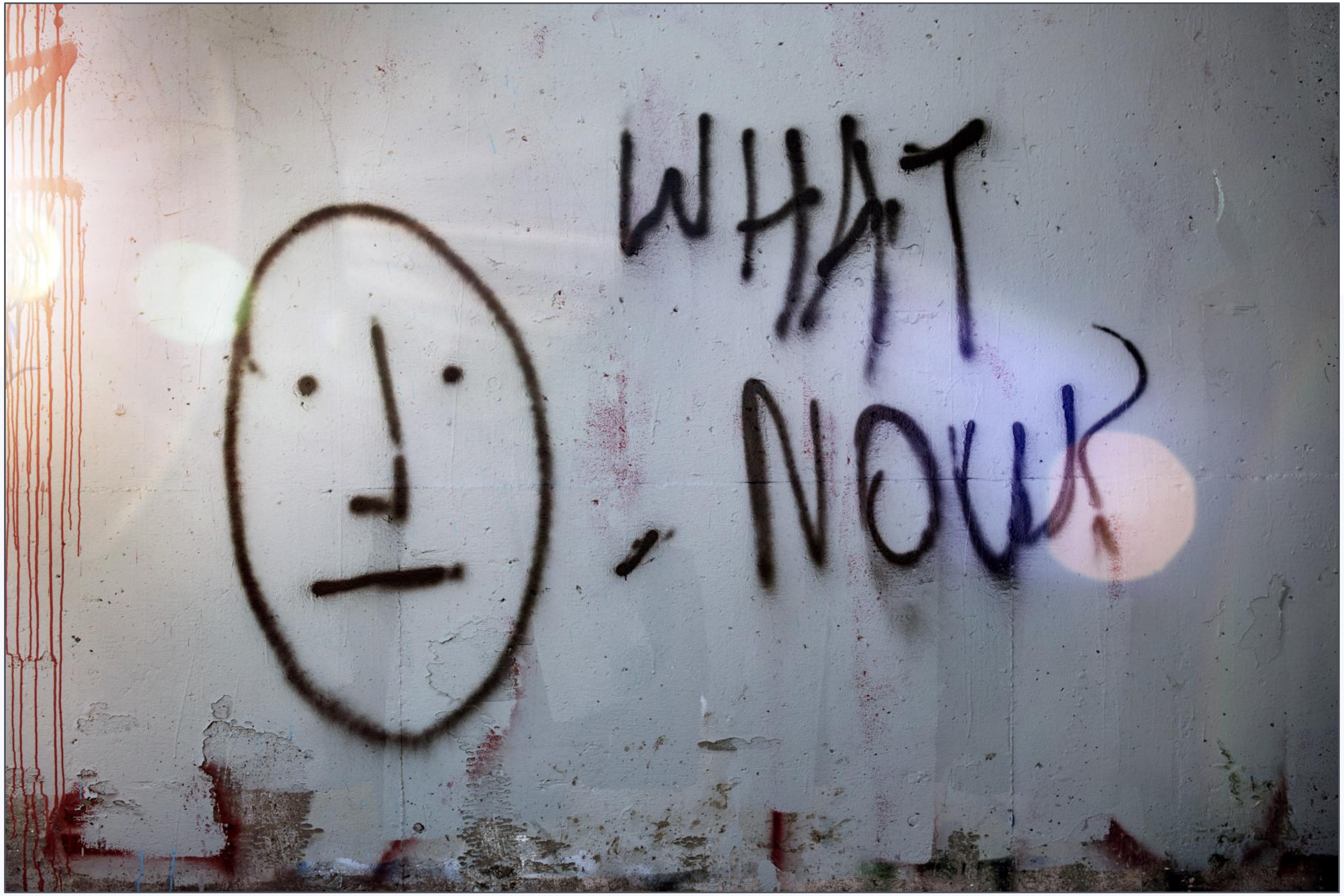


Definition of "*technical debt*":

In software development, technical debt […] is the **implied cost of future reworking required** when choosing an easy but **limited** solution instead of a better approach that could take more time.

Source: *Wikipedia*

Find <u>your</u> right way to the Azure Cloud



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Define **what** to achieve using Cloud Services?

1. Cloud Strategy

Know where you want to go in the Cloud

"Cloud can not deliver what it promises, if you do not know what to expect from it." 🗺️ 🛫

🎯 A well-thought-out **cloud strategy** is crucial to fully exploit technological **potential**. 🎯

Enterprise (Public) Cloud – Strategy

- Procedure & content of cloud strategy definition →

Cloud Strategy
(Business Output, Goals, Big Picture, etc.)

1. **Analysis of documentations & Feedback**
2. **Workshop** („*Status quo*“)
3. **Summary of current status**
4. **Workshop Goals** (Cloud specific)
5. **Strategy Paper** („*Big Picture*“)
6. **Presentation & Go (C-Level)**

Cloud Strategy – Overall:

- Company **overall** strategy
- Align organization’s **goals**
- Stakeholder identification
- Obstacles & Dependencies
- Time frame & Milestones
- Internal & external prerequisites

Cloud Strategy – Details:

- Cloud first, Cloud native, Cloud smart, etc.
- **Multi-cloud** scenarios (hyperscaler, platform comparison, exit strategy, vendor lock-in, etc.)
- Governance/Compliance (law, rules, audits, etc.)
- Cloud automation (IaC)
- Application Migration Strategy
- Application Modernization (Refactor, Rehost, Relocate, Repurchase, Retire, Retain, etc.)
- Organization (skillset, staffing, business processes, CCoE)

1. Cloud Strategy

Why to cover different areas of cloud management on Azure?

2. Azure Governance

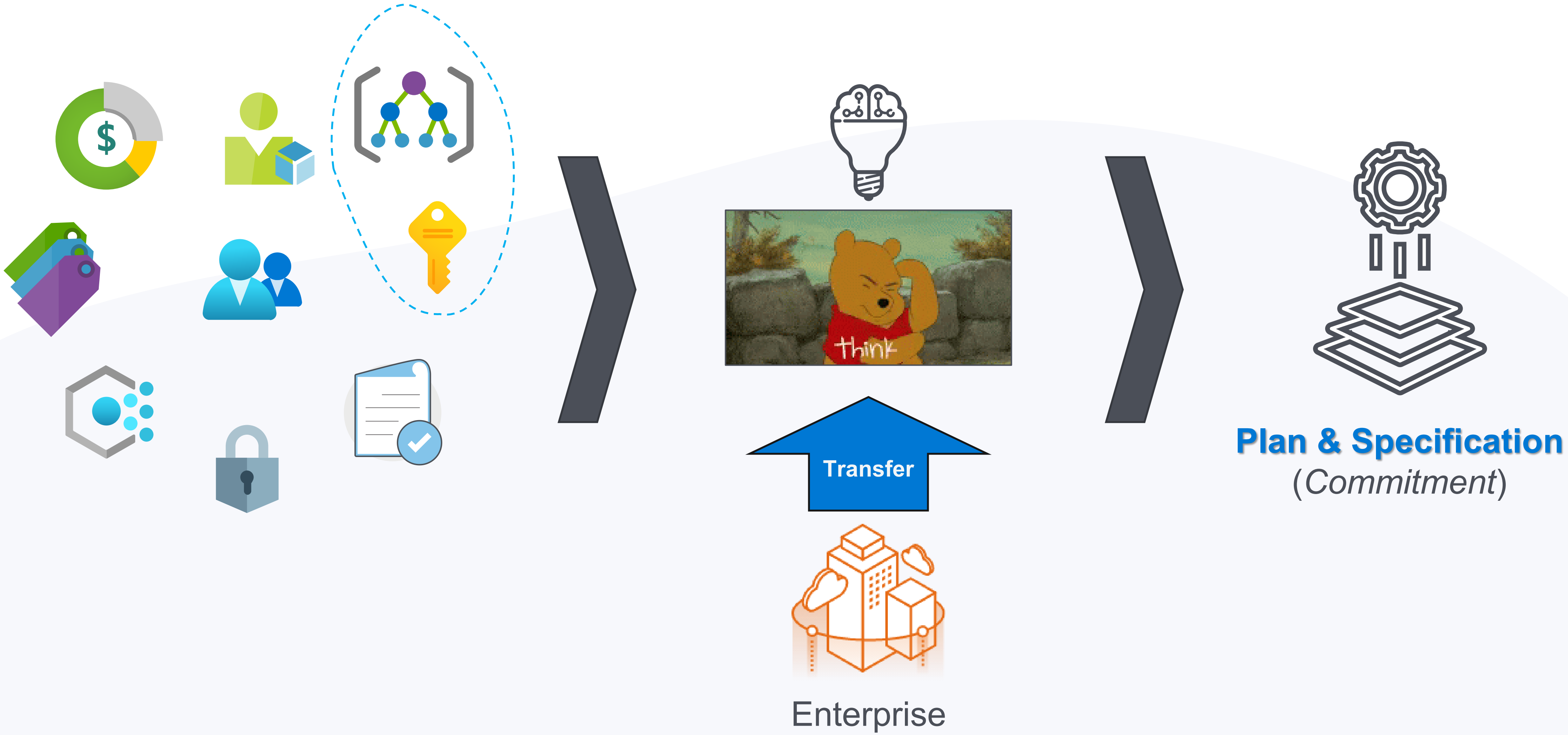
Azure Governance Topics & Capabilities

Why is a Governance Team needed?

Define the
"Rules of the Game"

- The official guidance in the CAF is to **always** create a cloud **governance team**. At first, the team might be very **small**.
- Ensures that cloud-adoption **risks** and risk tolerance are properly **evaluated** and **managed**.
- Identifies risks that **can't be tolerated by the business**, and it converts risks into governing corporate policies. (Link)

Azure Governance – “Tailor” all design areas before

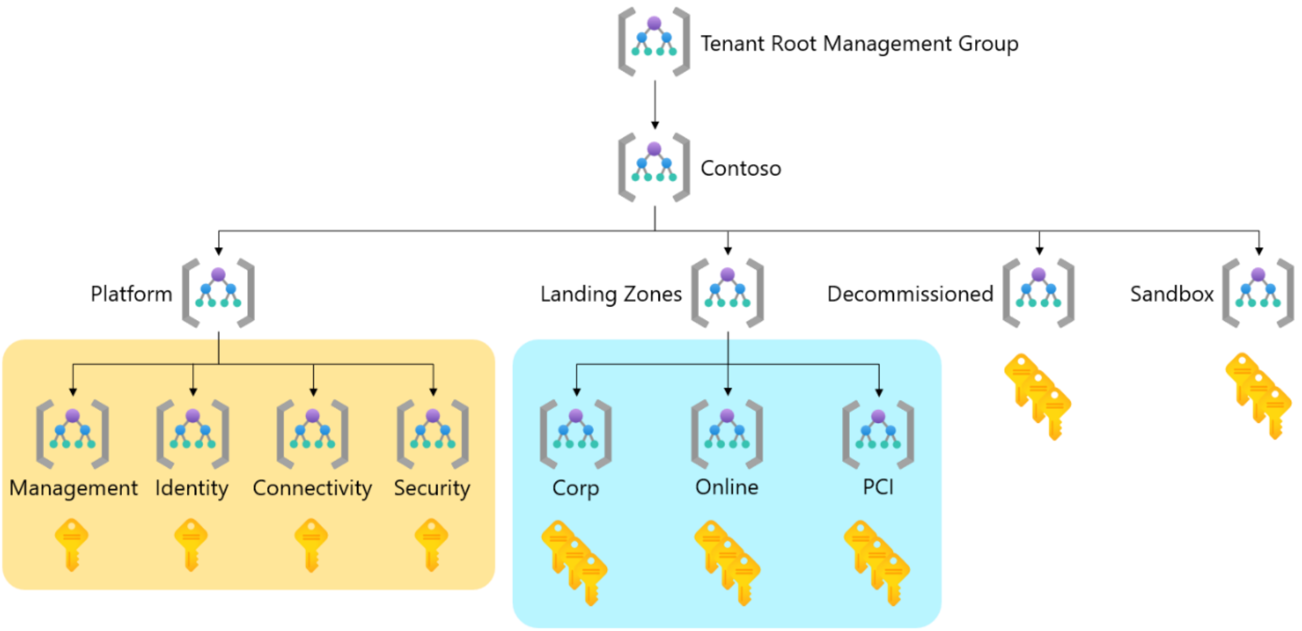


CAF – Azure **Hierarchy** – “*Tailored Example*”

Link: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/tailoring-alz#example-of-a-tailored-azure-landing-zone-hierarchy

Example of a tailored Azure landing zone hierarchy

The following diagram shows a tailored Azure landing zone hierarchy. It uses examples from the preceding diagram.



2. Azure Governance

Describe your basic infrastructure components on Azure.

3.1 Azure Core Infrastructure

Why the Microsoft CAF is not enough?

*“A **framework** is a generic term commonly referring to an essential supporting structure which other things are built on top of.”*

(Source: Framework – Wikipedia)

Framework = Construction Kit

Customer Requirements

Microsoft CAF for Azure

Know-How
Design
Implementation

Cloud Adoption **Templates**
&
Standardized **Blueprints**

…for Azure

DEMO Time

Azure Core Infrastructure

Azure Basic/Core General Design

- **Requirements** of the core layer (“*Landing Zone*”).
- Alignment of Azure **Governance** & **Architecture**.
- Design overall **network topology** in Azure (“on-premise to Azure”).
- Consideration **SLAs** and "*single point of failure*“ (SPOF).
- **Responsibilities** for the core layer (interdisciplinary teams & CCoE).
- Azure **Regions**, Availability **Zones** & Region **Pairs**.
- Implementation of all core/basic resources with **IaC** approach (*Terraform*).
- Definition of IT Cloud **Operations** & **Managed Services.**

Azure Network Architecture

- Analysis of detailed network requirements (hub & spoke, latency, vWAN, etc.).
- Address range definition and design (CIDR, VNets, Subnets, Peerings, etc.).
- Network **segmentation** & security **zones**.
- Design of a **routing concept** including standardized **traffic flow.**
- **DNS** resolution (possibility to use VMs or Azure DNS Private Resolver).
- **Process** for network adjustments, troubleshooting and expansion requests.
- **Mapping** of the Azure Network in a standardized **IaC library** (e.g., Terraform).

Azure Hybrid Connection

- Analysis of the **local** (*on-premises*) data centers as a starting point.
- Detailed connection **requirements** (bandwidth, outbound traffic, etc.).
- Define **shared services** in the hub network (as the first "*landing zone*").
- Alignment Azure network and hybrid connection (intersection).
- Costs & Definition of a roadmap for hybrid connections (VPN vs. ExpressRoute).
- **Mapping** of the Azure Network in a standardized **IaC library** (e.g., Terraform).

Azure Traffic Control (Firewall & NVA)

- Requirements for an **Azure NVA** (IaaS) or **Azure Firewall** (PaaS).
- Definition of (high) **availability** for the firewall solution in Azure.
- Avoidance of a single point of failure (SPOF) because of **redundancy.**
- Design firewall **architecture** (VNETs, CIDR blocks, Public IPs, Forced Tunneling, etc.).
- Challenge of the **routing concept** from the Azure Network (*traffic flow*).
- Definition needed **features** (e.g., Proxy, Threat Int., IDPS, TLS inspections, etc.).
- Template <u>standardized</u> settings & rule sets (FW Policies – RCG with NAT, NET, APP rules).
- Deployment of the Firewall solution with **IaC library** (e.g., Terraform).

Logging & Monitoring

- Requirements regarding a **monitoring solution** for Azure resources.
- Application/solution monitoring, infrastructure monitoring.
- Specification of the KPI for Logs & Metrics in Azure (CPU, RAM, Disk, Availability, etc.).
- Use of current tools in combination with a monitoring concept (alerting pipeline).
- Configuration of the diagnostic settings for detailed analysis & monitoring.
- Creation of a dedicated concept for mapping a monitoring strategy.

3.1 Azure Core Infrastructure

Describe other needed infrastructure components on Azure.

3.2 Azure Extended Infrastructure

Extended Design Areas & Specifications

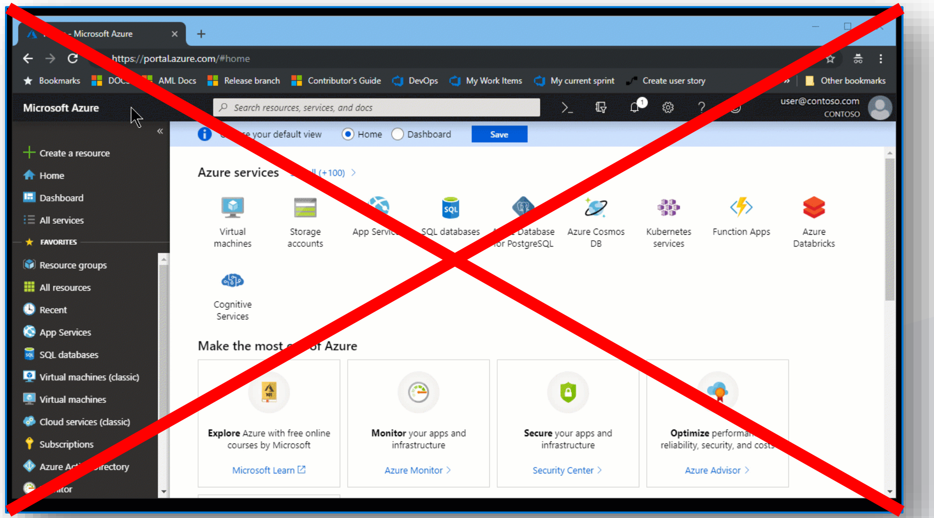
- **Backup**-Strategy (on-premise, Azure Backup, etc.).
- Azure Site Recovery (**BCDR**).
- **“Core”**-Services (e.g.: DC, DNS, Jumphosts, Bastion, Firewall, etc.).
- Key-/Secret- & Certificate-Management (**KSC**).
- **Network Security** (e.g.: Internet-Breakout, Stages, Endpoints).
- **Serverbereitstellung** (e.g.: VMs, Images, Agents, SKUs, Update, ALM, Encryption, etc.).
- Microsoft Defender for Cloud (*Azure Security Center*).
- Orientation: Azure Reference Architecture (Link) & well-architected Framework (Link)

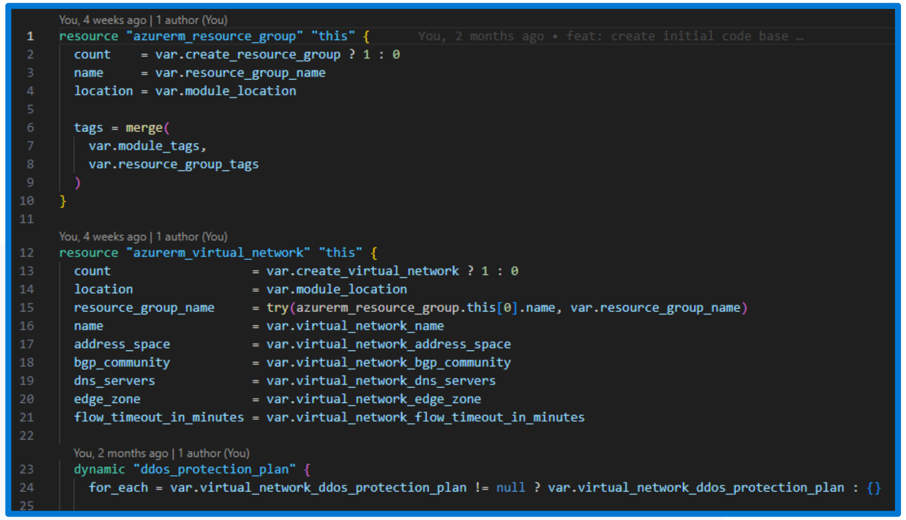
3.2 Azure Extended Infrastructure

Why focus on IaC instead of Click-Ops?

4. Cloud Automation

Mind change of (*Cloud* ☁️) resource deployment





```
You, 4 weeks ago | 1 author (You)
resource "azurerm_resource_group" "this" {          You, 2 months ago • feat: create initial code base …
  count    = var.create_resource_group ? 1 : 0
  name     = var.resource_group_name
  location = var.module_location

  tags = merge(
    var.module_tags,
    var.resource_group_tags
  )
}

You, 4 weeks ago | 1 author (You)
resource "azurerm_virtual_network" "this" {
  count                   = var.create_virtual_network ? 1 : 0
  location                = var.module_location
  resource_group_name     = try(azurerm_resource_group.this[0].name, var.resource_group_name)
  name                    = var.virtual_network_name
  address_space           = var.virtual_network_address_space
  bgp_community           = var.virtual_network_bgp_community
  dns_servers             = var.virtual_network_dns_servers
  edge_zone               = var.virtual_network_edge_zone
  flow_timeout_in_minutes = var.virtual_network_flow_timeout_in_minutes

  You, 2 months ago | 1 author (You)
  dynamic "ddos_protection_plan" {
    for_each = var.virtual_network_ddos_protection_plan != null ? var.virtual_network_ddos_protection_plan : {}
```

„Click-Click-Cloud"
„Click-Ops" 🖱️
„Clicky-Bunti"

🤔 →

<u>Mind Change:</u> 🧠
From "*static*" to
"*dynamic*" infrastructure

💡 →

CI/CD-driven
Infrastructure deployment ⚙️

What is IaC?

Provisioning & Management of infrastructures using **code**.

What resources, applications and environments are **configured**.

Declarative syntax can be used to specify the resources and settings.

Code describes the resources and the whole **architecture** landscapes.

Humanreadable code files (`.tfvars`) describe how the infrastructure looks like.

Definition of **variables** and their **values** per environment.

DEMO Time

Landing Zone Deployment

4. Cloud Automation (“IaC”)

Modularization as IaC gamechanger.

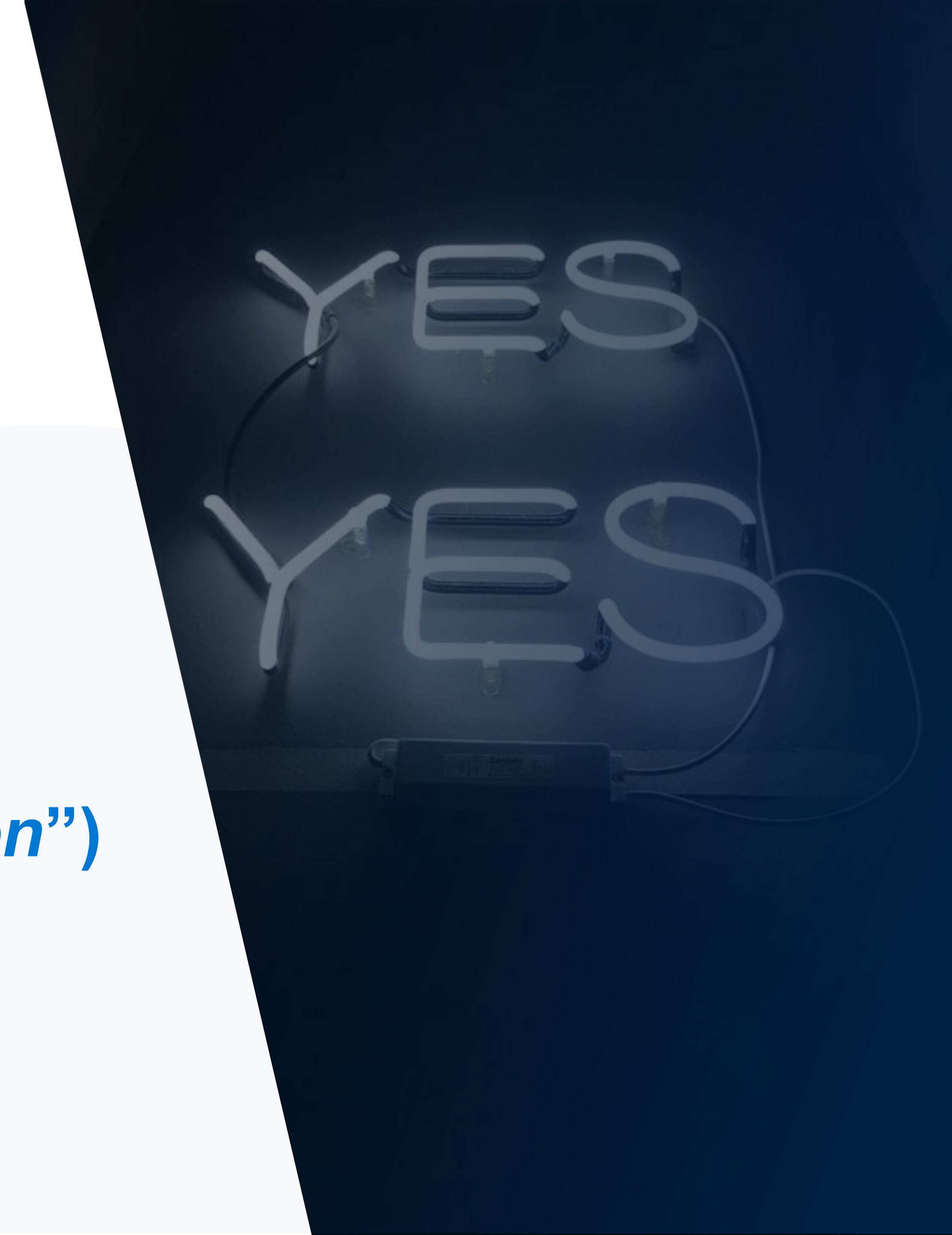
5. Module Library

What is a module?

- Create lightweight **abstractions**
- Don't describe **physical** objects
- Describe infrastructure in terms of its **architecture**
- Package and **reuse** resource configurations
- Separated in code **repositories**
- **Independently** usable & testable

"A module is a container for multiple resources that are used together."

4. Cloud Automation (“*Modularization*”)

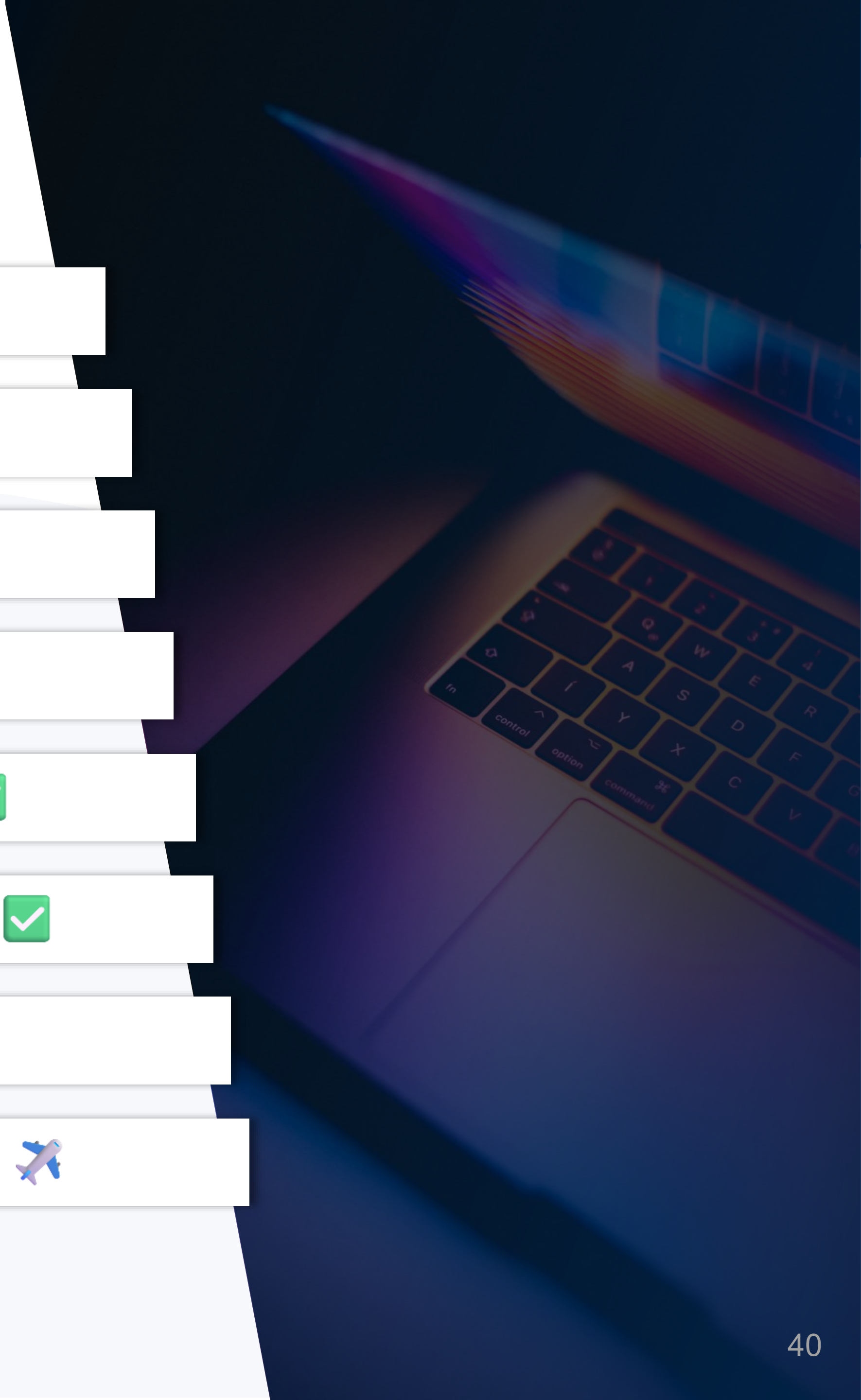


What are the essential take-aways?

5. Summary

Conclusion – Design Principals

Know your goals/destination	→ **Cloud Strategy** ✅
Don’t do things “later”	→ **Technical Dept** ✅
Define the rules & guardrails	→ **Governance** Concept & Team ✅
Get orientation & rethink	→ Microsoft **CAF** for Azure ✅
Design & Architecture	→ **Core** Infrastructure (“*Extended*”) ✅
Modularization	→ Provide shared IaC **Module Library** ✅
Don’t do “*Click-Click-Cloud*”	→ Infrastructure as Code – **IaC** ✅
Stay on Track	→ Follow your planned **Cloud Journey** ✈️



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Q & A